

Sequence design ID	Mutations	pH	Conversion [%]	EE [%]
WT (Gre2)	—	4.0	7.2	5.0
WT (Gre2)	—	7.0	16.5	17.0
WT (Gre2)	—	10.0	18.3	20.4
1	T41S, F85L, H109E, M272T	4.0	16.8	19.8
1	T41S, F85L, H109E, M272T	7.0	29.2	37.8
1	T41S, F85L, H109E, M272T	10.0	39.0	60.5
3	F85L, S92P, R184K, M272T	4.0	18.3	21.9
3	F85L, S92P, R184K, M272T	7.0	25.8	31.4
3	F85L, S92P, R184K, M272T	10.0	37.8	58.0
7	N39Q, F85L, H109E, A152S, N289K, R341K	4.0	20.3	25.0
7	N39Q, F85L, H109E, A152S, N289K, R341K	7.0	26.8	31.8
7	N39Q, F85L, H109E, A152S, N289K, R341K	10.0	42.9	71.6
8	F85L, H109E, K174R, K207E, K311S, F338Y	4.0	10.7	9.0
8	F85L, H109E, K174R, K207E, K311S, F338Y	7.0	37.3	51.6
8	F85L, H109E, K174R, K207E, K311S, F338Y	10.0	26.8	33.0
17	M1G, F85L, K174E	4.0	17.5	20.9
17	M1G, F85L, K174E	7.0	28.4	35.8
17	M1G, F85L, K174E	10.0	39.3	61.1